

3.2 Hydrocarbons

Question Paper

Course	CIE A Level Chemistry
Section	3. Organic Chemistry
Topic	3.2 Hydrocarbons
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

Which equation represents a valid propagation step in the free radical reaction between ethane and chlorine?

- A. $\text{C}_2\text{H}_6 + \text{Cl}\cdot \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{H}\cdot$
- B. $\text{C}_2\text{H}_5\text{Cl} + \text{Cl}\cdot \rightarrow \text{C}_2\text{H}_4\text{Cl}\cdot + \text{HCl}$
- C. $\text{C}_2\text{H}_6 + \text{H}\cdot \rightarrow \text{C}_2\text{H}_5\cdot + \text{HCl}$
- D. $\text{C}_2\text{H}_5\cdot + \text{Cl}\cdot \rightarrow \text{C}_2\text{H}_5\text{Cl}$

[1 mark]

Question 2

A fraction of distilled crude oil contains molecules with between 15 and 19 carbon atoms. This fraction is cracked by strong heating.

Why is this done?

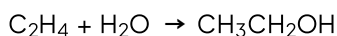
- 1 To produce alkenes.
- 2 To produce smaller molecules which are in higher demand.
- 3 To insert oxygen atoms into the hydrocarbons.

- A. 1, 2 and 3
- B. 1 and 2
- C. 2 and 3
- D. 1 only

[1 mark]

Question 3

Ethene reacts with steam in the presence of sulfuric acid.



What type of reaction is this?

- A. Acid / Base
- B. Addition
- C. Hydrolysis
- D. Substitution

[1 mark]

Question 4

Which statements about alkenes are correct?

- 1 They are less reactive than alkanes towards electrophiles.
- 2 They are used as monomers for polymerisation.
- 3 They are formed when higher alkanes are cracked.

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

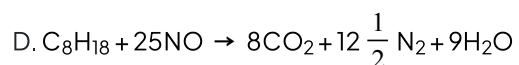
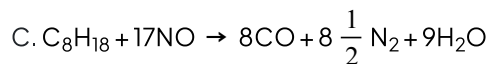
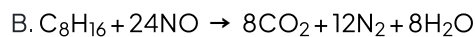
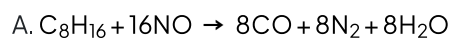
D. 3 only

[1 mark]

Question 5

One of the reactions taking place in a catalytic converter in a car exhaust system is between nitrogen oxide and octane (unburned petrol). The products of this reaction are non-toxic.

Which is the correct equation for the reaction?



[1 mark]

Question 6

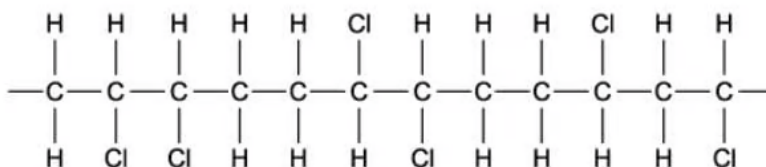
When bromine reacts with propene in an organic solvent at room temperature, what is the mechanism by which the bromine attacks the propene?

- A. Electrophilic addition
- B. Electrophilic substitution
- C. Nucleophilic addition
- D. Nucleophilic substitution

[1 mark]

Question 7

A molecule of a polymer contained the sequence shown.



Which monomer could produce this polymer by addition polymerisation?

- A. $\text{CHCl}=\text{CHCl}$
- B. $\text{CH}_2=\text{CHCl}$
- C. $\text{CH}_3\text{CCl}=\text{CHCl}$
- D. $\text{CH}_3\text{CCl}=\text{CH}_2$

[1 mark]

Question 8

Four students, W, X, Y and Z, made the following statements about alkanes and alkenes.

- W 'bromine reacts with alkanes by electrophilic substitution'
- X 'bromine reacts with alkenes by a free-radical addition mechanism'
- Y 'alkenes can be oxidised by acidified manganate(VII) ions'
- Z 'alkenes are formed from alkanes by cracking'

Which two students are correct?

- A. W and X
- B. W and Z
- C. X and Y
- D. Y and Z

[1 mark]

Question 9

The hydrocarbon $C_{17}H_{36}$ can be cracked.

Which compound is the **least** likely to be produced in this reaction?

- A. C_3H_8
- B. C_4H_8
- C. C_8H_{16}
- D. $C_{16}H_{34}$

[1 mark]

Question 10

In the presence of ultraviolet light, ethane and chlorine react to give a mixture of products.

Which compound could be present in the mixture of products?

- A. CH_3Cl
- B. $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$
- C. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
- D. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

[1 mark]

Model Answers: Easy

1

The correct answer is **B** because:

- Ethane reacts with chlorine by free radical halogenation in the presence of sunlight:
- Initiation step: $\text{Cl}_2 \rightarrow 2\text{Cl}\cdot$
 - Initiation = net formation of radicals
- Propagation step: $\text{C}_2\text{H}_6 + \text{Cl}\cdot \rightarrow \text{C}_2\text{H}_5\cdot + \text{HCl}$
- Propagation step: $\text{C}_2\text{H}_5\cdot + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{Cl}\cdot$
 - Propagation = no change in the number of free radicals
- Termination step: $\text{C}_2\text{H}_5\cdot + \text{Cl}\cdot \rightarrow \text{C}_2\text{H}_5\text{Cl}$
 - Termination = net destruction of free radicals
- Therefore, option **B** is the only option which shows a valid propagation step.

A & C are incorrect as there are no hydrogen radicals in the free radical reaction between ethane and chlorine.

D is incorrect as this shows a net destruction of free radicals which could be a termination step.

2

The correct answer is **B** because:

- Cracking is a reaction in which the larger saturated hydrocarbon molecules in crude oil are broken down into smaller, more useful hydrocarbon molecules.
 - Therefore, statement 2 is correct.
- Some of the molecules produced from cracking are unsaturated hydrocarbons and hence why this process is used to produce alkenes.

- Therefore, statement 1 is correct.

A & C are incorrect as cracking simply breaks long chain hydrocarbons into smaller molecules, so, no oxygen atoms are added.

D is incorrect as cracking produces both short chain alkenes and alkanes, both of which are in higher demand than longer chain hydrocarbons.

3

The correct answer is **B** because:

- In an addition reaction, two compounds come together to form one larger compound.
- There are no other products produced.
- Therefore, the reaction between ethene and steam is an addition reaction.

A is incorrect as an acid-base reaction involves an acid and a base as reactants.

C is incorrect as in a hydrolysis reaction, chemical bonds are broken by the addition of water, however that is not the case here.

D is incorrect as in a substitution reaction, an atom or functional group is replaced by another similar species.

4

The correct answer is **C** because:

- Under the right conditions, alkenes can be used as monomers for addition polymerisation to make long chain saturated polymers.
 - Therefore, statement 2 is correct.
- Cracking is a reaction in which the larger saturated hydrocarbon molecules (alkanes) in crude oil are broken down into smaller, more useful hydrocarbon molecules, including both alkenes and alkanes.
 - Therefore, statement 3 is correct.

A & B are incorrect as alkenes are **more** reactive than alkanes as the double bond is

more easily able to open up towards electrophiles.

D is incorrect as statement 3 is not the only correct statement.

5

The correct answer is **D** because:

- C_8H_{18} is the correct formula for octane.
- Carbon dioxide (CO_2), nitrogen gas (N_2) and water (H_2O) are all non-toxic products.

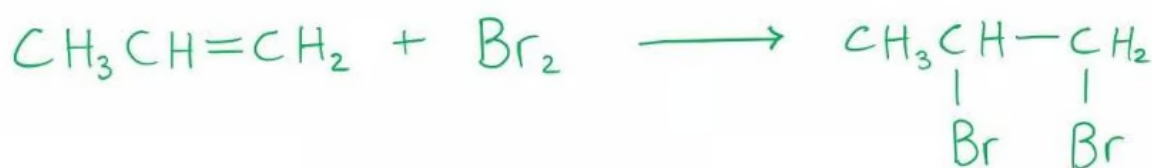
A & **B** are incorrect as octane is an alkane with formula C_8H_{18} , whereas **A** and **B** show C_8H_{16} , which is octene.

C is incorrect as CO is carbon monoxide, which is a toxic gas.

6

The correct answer is **A** because:

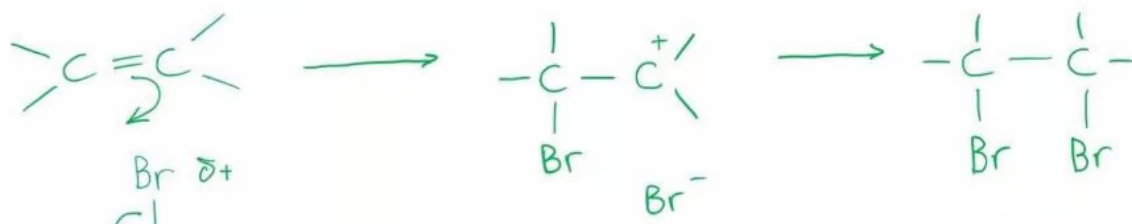
- In an addition reaction, two compounds come together to form one larger compound.
- There are no other products produced, as shown:



- In the reaction between propene and bromine:
 - The positively charged bromine atom acts as an electrophile (electron loving species), so it reacts with the double carbon bond.
 - A pair of electrons from the carbon double bond moves onto the positive bromine atom.
 - The double bond breaks, and a bromine atom becomes attached to each carbon.

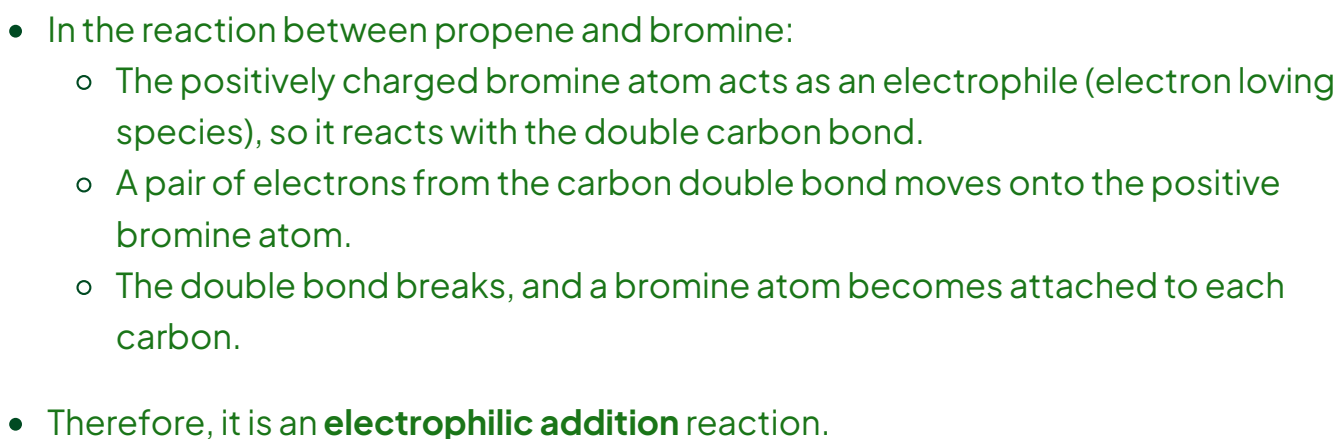
- Therefore, it is an **electrophilic addition** reaction.

Electrophilic addition mechanism:



6

- In an addition reaction, two compounds come together to form one larger compound.
- There are no other products produced, as shown:

[illegible]

7

The correct answer is **B** because:

Step	Working out
1. Identify the monomers in the displayed diagram	
2. Break the bond between these monomers	
3. Add in the double bond between the carbon atoms	
4. Convert the structural formula to the condensed formula	$\text{CH}_2 = \text{CHCl}$

8

The correct answer is **D** because:

- Cracking is a reaction in which the larger saturated hydrocarbon molecules (alkanes) in crude oil are broken down into smaller, more useful hydrocarbon molecules, including alkenes.
 - Therefore, statement Z is correct.
- The carbon-carbon double bond in an alkene reacts with potassium manganate(VII) solution – this is an oxidation reaction.

- Therefore, statement Y is correct.

A & B are incorrect as bromine does not react with alkanes due to their lack of double bond, so, statement X is incorrect.

C is incorrect as bromine reacts with alkenes via electrophilic addition, so, statement W is incorrect.

9

The correct answer is **D** because:

- Cracking is a reaction in which the larger saturated hydrocarbon molecules (alkanes) in crude oil are broken down into smaller, more useful hydrocarbon molecules, including both alkanes and alkenes.
- Typically, in cracking reactions, a mixture of different alkanes and alkenes are produced.
- So, for the hydrocarbon $C_{17}H_{36}$, the cracking process is likely to produce alkanes and alkenes of less than 10 carbons.
- Therefore, the compound **least** likely to be produced in this reaction is $C_{16}H_{34}$

10

The correct answer is **C** because:

- In the initiation step, free radical $C/•$ atoms are formed:
- Initiation step: $Cl_2 \rightarrow 2C/•$
 - Initiation = net formation of radicals
- These free radical $C/•$ atoms remove a hydrogen from the ethane making an ethyl free radical, $C_2H_5•$:
- Propagation step: $C_2H_6 + C/• \rightarrow C_2H_5• + HC/•$
 - Propagation = no change in the number of free radicals
- Two of these radicals can then pair-up in a termination step to form butane $CH_3CH_2CH_2CH_3$ (or C_4H_{10}):

- Termination step: $\text{C}_2\text{H}_5\cdot + \text{C}_2\text{H}_5\cdot \rightarrow \text{C}_4\text{H}_{10}$
 - Termination = net destruction of free radicals
- Therefore, option **C** is the only option which shows a possible compound that could be present.

